Machine Learning Canvas

	PREDICTIONS	OBJECTIVES	DATA
IDEA	**Context** Who uses/is affected? • Retention, Customer Success, Marketing, Operations, and leadership use the score. • Customers may receive retention offers or service follow-up.	**Value proposition** What are we trying to do? • Reduce avoidable churn and protect recurring revenue. • Prioritize limited retention capacity consistently through an API.	**Data sources** Where does data come from? • Telco churn XLSX workbook and generated encoded CSV. • Future: customer, billing, service, and outcome systems.
SPECS	**Problem** Question to predict answers to Will this customer churn without retention action? Input 24 raw fields: profile, location, tenure, services, contract, billing, and CLTV. Outputs Churn / No churn plus probability. Type Binary classification. **Baseline** Majority class and simple business rules based on tenure, contract, and charges.	**Performance evaluation** Production / business metrics • Retention uplift versus control. • Churn-rate reduction. • Cost per retained customer and ROI. Prediction metrics • Primary: F1. • Supporting: ROC AUC, balanced accuracy, precision, recall, accuracy. • Monitor false negatives and false positives. Offline method Stratified 80/20 split plus five-fold StratifiedKFold on training data; select by CV F1.	**Dataset** Collection and size 7,043 records and 33 source columns from the Telco churn workbook. Historical churn is the supervised outcome. Data and models are versioned with DVC. **Features** Inputs extracted from the source • Profile, geography, tenure, services, contract, billing, and CLTV. • One-hot encoding for categoricals; frequency encoding for City. • 35 encoded predictors. Excluded: Churn Score, Churn Label, Churn Reason, and other target-derived fields.
DEPLOYMENT	**Using predictions** When/how many? On-demand single-customer scoring through POST /predict; batch scoring can be added for campaigns. Time constraint Normal web-request latency for one record. How used? • Rank customers for human review. • Apply a business-approved threshold. • Combine risk with customer value and campaign capacity. • Swagger /docs supports synthetic Try it out requests.	**Learning predictive models** When/update data? Retrain with new outcomes or detected drift; use time-aware historical snapshots when possible. Training constraint Offline notebook workflow; production retraining should be scheduled and released under review. Deployment criteria • Beat baseline on F1 and maintain recall. • Check ROC AUC, calibration, subgroup performance, and ROI. • Version preprocessing and artifact together. Champion: Random Forest, 300 trees, balanced classes; CV F1 0.6466 ± 0.0263, test F1 0.6315, ROC AUC 0.8465.	